We will now pick tags for $\mathcal{Q}$ that will provide a contradiction to (6). We tag $\mathcal{Q}$ by $t_i := x_i$ for $i \neq k$ and we pick $t_k \in [x_{k-1}, x_k]$ such that

$$|f(t_k)(x_k - x_{k-1})| > |L| + 1 + \left|\sum_{i \neq k} f(t_i)(x_i - x_{i-1})\right|.$$

From the Triangle Inequality (in the form $|A + B| \geq |A| - |B|$), we have

$$\left|S(f; \dot{\mathcal{Q}})\right| \geq |f(t_k)(x_k - x_{k-1})| - \left|\sum_{i \neq k} f(t_i)(x_i - x_{i-1})\right| > |L| + 1,$$

which contradicts (6). Q.E.D.

We will close this section with an example of a function that is discontinuous at every rational number and is not monotone, but is Riemann integrable nevertheless.

7.1.7 Example We consider Thomae's function $h : [0, 1] \to \mathbb{R}$ defined, as in Example 5.1.6(h), by $h(x) := 0$ if $x \in [0, 1]$ is irrational, $h(0) := 1$ and by $h(x) := 1/n$ if $x \in [0, 1]$ is the rational number $x = m/n$ where $m, n \in \mathbb{N}$ have no common integer factors except 1. It was seen in 5.1.6(h) that h is continuous at every irrational number and discontinuous at every rational number in [0, 1]. See Figure 5.1.2. We will now show that $h \in \mathcal{R}[0, 1]$.

For $\varepsilon > 0$, the set $E := \{x \in [0, 1] : h(x) \geq \varepsilon/2\}$ is a finite set. (For example, if $\varepsilon/2 = 1/5$, then there are eleven values of x such that $h(x) \geq 1/5$, namely, $E = \{0, 1, 1/2, 1/3, 2/3, 1/4, 3/4, 1/5, 2/5, 3/5, 4/5\}$. (Sketch a graph.) We let n be the number of elements in E and take $\delta := \varepsilon/(4n)$. If $\dot{\mathcal{P}}$ is a given tagged partition such that $||\dot{\mathcal{P}}|| < \delta$, then we separate $\dot{\mathcal{P}}$ into two subsets. We let $\dot{\mathcal{P}}_1$ be the collection of tagged intervals in $\dot{\mathcal{P}}$ that have their tags in E, and we let $\dot{\mathcal{P}}_2$ be the subset of tagged intervals in $\dot{\mathcal{P}}$ that have their tags elsewhere in [0, 1]. Allowing for the possibility that a tag of $\dot{\mathcal{P}}_1$ may be an endpoint of adjacent intervals, we see that $\dot{\mathcal{P}}_1$ has at most $2n$ intervals and the total length of these intervals can be at most $2n\delta = \varepsilon/2$. Also, we have $0 < h(t_i) \leq 1$ for each tag t_i in $\dot{\mathcal{P}}_1$. Consequently, we have $S(h; \dot{\mathcal{P}}_1) \leq 1 \cdot 2n\delta \leq \varepsilon/2$. For tags t_i in $\dot{\mathcal{P}}_2$, we have $h(t_i) < \varepsilon/2$ and the total length of the subintervals in $\dot{\mathcal{P}}_2$ is clearly less than 1, so that $S(h; \dot{\mathcal{P}}_2) < (\varepsilon/2) \cdot 1 = \varepsilon/2$. Therefore, combining these results, we get

$$0 \leq S(h; \dot{\mathcal{P}}) = S(h; \dot{\mathcal{P}}_1) + S(h; \dot{\mathcal{P}}_2) < \varepsilon/2 + \varepsilon/2 = \varepsilon.$$

Since $\varepsilon > 0$ is arbitrary, we infer that $h \in \mathcal{R}[0, 1]$ with integral 0. □

Exercises for Section 7.1

1. If $I := [0, 4]$, calculate the norms of the following partitions:
   (a) $\mathcal{P}_1 := (0, 1, 2, 4)$, (b) $\mathcal{P}_2 := (0, 2, 3, 4)$,
   (c) $\mathcal{P}_3 := (0, 1, 1.5, 2, 3.4, 4)$, (d) $\mathcal{P}_4 := (0, .5, 2.5, 3.5, 4)$.

2. If $f(x) := x^2$ for $x \in [0, 4]$, calculate the following Riemann sums, where $\dot{\mathcal{P}}_i$ has the same partition points as in Exercise 1, and the tags are selected as indicated.
   (a) $\dot{\mathcal{P}}_1$ with the tags at the left endpoints of the subintervals.
   (b) $\dot{\mathcal{P}}_1$ with the tags at the right endpoints of the subintervals.
   (c) $\dot{\mathcal{P}}_2$ with the tags at the left endpoints of the subintervals.
   (d) $\dot{\mathcal{P}}_2$ with the tags at the right endpoints of the subintervals.

3. Show that $f : [a, b] \to \mathbb{R}$ is Riemann integrable on $[a, b]$ if and only if there exists $L \in \mathbb{R}$ such that for every $\varepsilon > 0$ there exists $\delta_\varepsilon > 0$ such that if $\dot{\mathcal{P}}$ is any tagged partition with norm $||\dot{\mathcal{P}}|| \leq \delta_\varepsilon$, then $\left|S(f; \dot{\mathcal{P}}) - L\right| \leq \varepsilon$.

4. Let $\dot{\mathcal{P}}$ be a tagged partition of $[0, 3]$.
   (a) Show that the union U_1 of all subintervals in $\dot{\mathcal{P}}$ with tags in $[0, 1]$ satisfies $\left[0, 1 - \|\dot{\mathcal{P}}\|\right] \subseteq U_1 \subseteq \left[0, 1 + \|\dot{\mathcal{P}}\|\right]$.
   (b) Show that the union U_2 of all subintervals in $\dot{\mathcal{P}}$ with tags in $[1,2]$ satisfies $\left[1 + \|\dot{\mathcal{P}}\|, 2 - \|\dot{\mathcal{P}}\|\right] \subseteq U_2 \subseteq \left[1 - \|\dot{\mathcal{P}}\|, 2 + \|\dot{\mathcal{P}}\|\right]$.
5. Let $\dot{\mathcal{P}} := \{(I_i, t_i)\}_{i=1}^n$ be a tagged partition of $[a, b]$ and let $c_1 < c_2$.
   (a) If u belongs to a subinterval I_i whose tag satisfies $c_1 \leq t_i \leq c_2$, show that $c_1 - \|\dot{\mathcal{P}}\| \leq u \leq c_2 + \|\dot{\mathcal{P}}\|$.
   (b) If $v \in [a, b]$ and satisfies $c_1 + \|\dot{\mathcal{P}}\| \leq v \leq c_2 - \|\dot{\mathcal{P}}\|$, then the tag t_i of any subinterval I_i that contains v satisfies $t_i \in [c_1, c_2]$.
6. (a) Let $f(x) := 2$ if $0 \leq x < 1$ and $f(x) := 1$ if $1 \leq x \leq 2$. Show that $f \in \mathcal{R}[0, 2]$ and evaluate its integral.
   (b) Let $h(x) := 2$ if $0 \leq x < 1$, $h(1) := 3$ and $h(x) := 1$ if $1 < x \leq 2$. Show that $h \in \mathcal{R}[0, 2]$ and evaluate its integral.
7. Use Mathematical Induction and Theorem 7.1.5 to show that if $f_1, \ldots, f_n$ are in $\mathcal{R}[a, b]$ and if $k_1, \ldots, k_n \in \mathbb{R}$, then the linear combination $f = \sum_{i=1}^n k_i f_i$ belongs to $\mathcal{R}[a, b]$ and $\int_a^b f = \sum_{i=1}^n k_i \int_a^b f_i$.
8. If $f \in \mathcal{R}[a, b]$ and $|f(x)| \leq M$ for all $x \in [a, b]$, show that $\left|\int_a^b f\right| \leq M(b - a)$.
9. If $f \in \mathcal{R}[a, b]$ and if $(\dot{\mathcal{P}}_n)$ is any sequence of tagged partitions of $[a, b]$ such that $\|\dot{\mathcal{P}}_n\| \to 0$, prove that $\int_a^b f = \lim_n S(f; \dot{\mathcal{P}}_n)$.
10. Let $g(x) := 0$ if $x \in [0, 1]$ is rational and $g(x) := 1/x$ if $x \in [0, 1]$ is irrational. Explain why $g \notin \mathcal{R}[0, 1]$. However, show that there exists a sequence $(\dot{\mathcal{P}}_n)$ of tagged partitions of $[a, b]$ such that $\|\dot{\mathcal{P}}_n\| \to 0$ and $\lim_n S(g; \dot{\mathcal{P}}_n)$ exists.
11. Suppose that f is bounded on $[a, b]$ and that there exists two sequences of tagged partitions of $[a, b]$ such that $\|\dot{\mathcal{P}}_n\| \to 0$ and $\|\dot{\mathcal{Q}}_n\| \to 0$, but such that $\lim_n S(f; \dot{\mathcal{P}}_n) \neq \lim_n S(f; \dot{\mathcal{Q}}_n)$. Show that f is not in $\mathcal{R}[a, b]$.
12. Consider the Dirichlet function, introduced in Example 5.1.6(g), defined by $f(x) := 1$ for $x \in [0, 1]$ rational and $f(x) := 0$ for $x \in [0, 1]$ irrational. Use the preceding exercise to show that f is *not* Riemann integrable on $[0, 1]$.
13. Suppose that $c \leq d$ are points in $[a, b]$. If $\varphi : [a, b] \to \mathbb{R}$ satisfies $\varphi(x) = \alpha > 0$ for $x \in [c, d]$ and $\varphi(x) = 0$ elsewhere in $[a, b]$, prove that $\varphi \in \mathcal{R}[a, b]$ and that $\int_a^b \varphi = \alpha(d - c)$. [*Hint*: Given $\varepsilon > 0$ let $\delta_\varepsilon := \varepsilon/4\alpha$ and show that if $\|\dot{\mathcal{P}}\| < \delta_\varepsilon$ then we have $\alpha(d - c - 2\delta_\varepsilon) \leq S(\varphi; \dot{\mathcal{P}}) \leq \alpha(d - c + 2\delta_\varepsilon)$.]
14. Let $0 \leq a < b$, let $Q(x) := x^2$ for $x \in [a, b]$ and let $\mathcal{P} := \{[x_{i-1}, x_i]\}_{i=1}^n$ be a partition of $[a, b]$. For each i, let q_i be the positive *square root* of
$$\tfrac{1}{3}\left(x_i^2 + x_i x_{i-1} + x_{i-1}^2\right).$$
   (a) Show that q_i satisfies $0 \leq x_{i-1} \leq q_i \leq x_i$.
   (b) Show that $Q(q_i)(x_i - x_{i-1}) = \frac{1}{3}\left(x_i^3 - x_{i-1}^3\right)$.
   (c) If $\dot{\mathcal{Q}}$ is the tagged partition with the same subintervals as $\mathcal{P}$ and the tags q_i, show that $S(Q; \dot{\mathcal{Q}}) = \frac{1}{3}\left(b^3 - a^3\right)$.
   (d) Use the argument in Example 7.1.4(c) to show that $Q \in \mathcal{R}[a, b]$ and
$$\int_a^b Q = \int_a^b x^2 dx = \tfrac{1}{3}\left(b^3 - a^3\right).$$
15. If $f \in \mathcal{R}[a, b]$ and $c \in \mathbb{R}$, we define g on $[a + c, b + c]$ by $g(y) := f(y - c)$. Prove that $g \in \mathcal{R}[a + c, b + c]$ and that $\int_{a+c}^{b+c} g = \int_a^b f$. The function g is called the c-**translate** of f.